

Real-time Obstacle Perception and Distance Estimation Based on Binocular Vision

1st Yaming Dong
College of Engineering
Yanbian University
Yanji, China
3105916652@qq.com

2nd Hua Jin*
College of Engineering
Yanbian University
Yanji, China
hjin@ybu.edu.cn

Abstract—To address the travel safety challenges faced by visually impaired individuals in complex environments, this study proposes a real-time obstacle perception and distance estimation method based on binocular vision. The system employs an improved YOLOv11n model to achieve high-precision obstacle detection (mAP@0.5: 82.83%) on the WOTR dataset, combined with a lightweight stereo matching network LightStereo for depth estimation. Through TensorRT FP16 quantization technology, the inference speed of object detection and stereo matching is improved by 3.2 times and 3.4 times respectively. The system is deployed on the NVIDIA Jetson Orin NX platform, capturing images in real-time through binocular cameras, obtaining obstacle categories and position distance information using object detection and stereo matching models, and outputting information through voice modules. The measured end-to-end inference latency is 46.76 ms, with absolute distance measurement error less than 0.25 meters within a 0-4 meter range, meeting the real-time and accuracy requirements for travel assistance for visually impaired individuals.

Keywords- Visual impairment assistance, Binocular vision, YOLOv11, Stereo matching, Embedded system, Model quantization

I. INTRODUCTION

According to the World Health Organization (WHO), there are 2.2 billion people worldwide with visual impairments. Visually impaired individuals face severe travel safety challenges, making accurate identification and distance measurement of environmental obstacles a critical technical problem for ensuring their independent mobility. Traditional white canes have sensing ranges less than 1.5 meters and cannot detect head-height obstacles; guide dogs require expensive training with extremely low coverage rates^[1]. Electronic Travel Aids (ETAs) provide environmental perception for visually impaired individuals through multimodal sensing technology, gradually becoming an emerging solution to overcome the limitations of traditional approaches.

Early ETAs were based on single sensors, providing only simple distance information and lacking semantic understanding capabilities. Breakthrough advances in computer vision technology, particularly the fusion of binocular stereo vision with deep learning, enable ETA systems to simultaneously acquire geometric information (position, size) and semantic attributes (category, state) of obstacles^[2]. Binocular stereo vision technology can simultaneously obtain geometric information and semantic attributes of obstacles^[3], offering better outdoor robustness, lower power consumption, and hardware costs compared to RGB-D cameras^[4]. However, existing binocular

vision systems still face challenges in visual impairment assistance, including insufficient real-time performance, limited accuracy, and difficulties in embedded deployment.

To address these issues, this study proposes a real-time obstacle perception and distance estimation method based on binocular vision, with the following main technical contributions:

- An improved YOLOv11n object detection model that achieves an mAP@0.5 of 82.83% on the WOTR dataset.
- Comparative analysis of multiple stereo matching algorithms demonstrates the rationality of selecting the LightStereo network for efficient and accurate depth estimation;
- Achieved 3.2× speedup for object detection model and 3.4× speedup for stereo matching model through TensorRT FP16 quantization;
- Integrated and deployed the entire system on NVIDIA Jetson Orin NX, achieving 46.76ms end-to-end inference latency with distance measurement error < 6.3% within 4 meters.

II. ALGORITHM SELECTION AND OPTIMIZATION

A. Object Detection Algorithm

Real-time high-precision obstacle recognition is fundamental for building reliable visual impairment assistance systems. This study adopts the YOLO (You Only Look Once) architecture as the core detection framework. As a typical single-stage detection algorithm, YOLO effectively balances accuracy and real-time requirements by converting the object detection task into a single regression problem^[5]. YOLOv11 is selected as the base architecture, with improvements in backbone network and feature fusion design enabling YOLOv11m to achieve higher mAP compared to YOLOv8m on the COCO dataset while reducing parameters by 22%^[6]. YOLOv11 demonstrates excellent cross-platform deployment adaptability, supporting cloud computing, edge devices, and NVIDIA GPU acceleration deployment scenarios, meeting the flexible deployment requirements of visual impairment assistance systems.

1) Dataset Construction

To adapt to the specific needs of travel assistance for visually impaired individuals, this study selected the public dataset WOTR^[7]. The dataset comprises 20 object categories relevant to pedestrian navigation scenarios, including: tree, red light, green

light, crosswalk, tactile paving, sign, pedestrian, bicycle, bus, truck, car, motorcycle, reflective cone, ashcan, warning column, roadblock, pole, dog, tricycle, and fire hydrant.

2) CSP-AWMSA Module

In travel scenarios for visually impaired individuals, the detection system needs to simultaneously process multiple types of targets with significant scale differences: guidance facilities, traffic participants, safety signs, and environmental landmarks. To address the limitation of fixed receptive fields in YOLOv1's original C3k2 structure (see Fig. 1) when processing cross-scale targets, a CSP Adaptive Weighted Multi-Scale Aggregation (CSP-AWMSA) module was designed to replace C3k2.

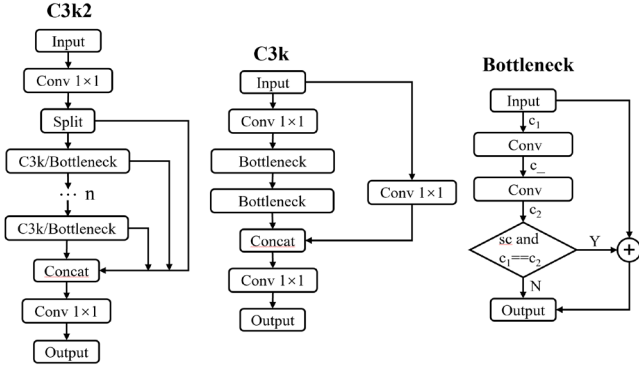


Fig. 1. C3k2 and its sub modules

The module contains two key components(see Fig. 2): (1) Progressive multi-scale feature aggregation unit, using cascaded 3×3 , 5×5 , 7×7 convolution kernels to build hierarchical receptive fields, achieving multi-level feature representation from local texture to global semantics through progressive expansion; (2) Channel attention-driven multi-branch weight generation mechanism, extracting channel statistical features through global adaptive pooling, constructing nonlinear mapping through two layers of 1×1 convolution and ReLU activation functions, and finally generating adaptive weights for three scale branches through Softmax function.

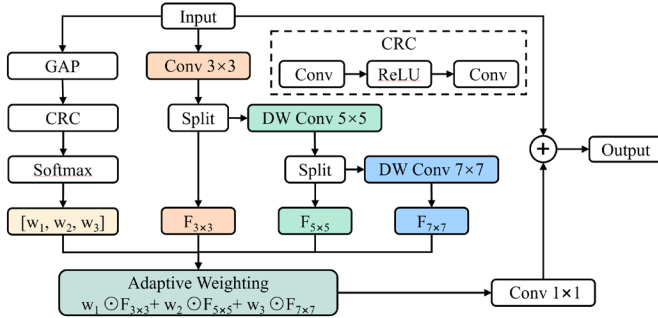


Fig. 2. CSP-AWMSA module

The training experiments were conducted on an NVIDIA A40 GPU with PyTorch 2.5.1 and CUDA 12.1. All models were trained for 300 epochs using standard optimization settings, as shown in TABLE I. Experimental results of the improved model on WOTR show that the CSP-AWMSA module significantly improves detection performance while maintaining parameter efficiency (parameter increment only 2.0%): average precision

mAP@0.5 improved from 80.90% to 82.83%, and mAP@0.5:0.95 increased from 55.98% to 57.55%, as shown in TABLE I, validating the effectiveness of the adaptive multi-scale feature aggregation strategy.

TABLE I. RESULTS OF MODEL TRAINING

Name	mAP50	mAP75	mAP50-95	Parameters	Size
Yolo11n	0.8090	0.6152	0.5598	2,585,467	5.2
CSP-AWMSA	0.8249	0.6288	0.5729	2,632,323	5.3

B. Stereo Matching Algorithm

Binocular stereo matching is the core technology for accurate distance estimation in our system. It computes the disparity between corresponding pixels in left and right images, from which depth is derived via triangulation.

Traditional methods like Block Matching (BM) and Semi-Global Block Matching (SGBM) are computationally efficient but struggle with textureless regions, lighting changes, and occlusions, often requiring extensive parameter tuning^[8]. In recent years, learning-based models (RAFT-Stereo^[9], IGEV-Stereo^[10]) have surpassed traditional methods in accuracy, but their large size and computational cost make them unsuitable for real-time embedded applications.

To achieve a balance between performance and efficiency, this system adopts the open-source LightStereo^[11] model as the core stereo matching algorithm. This algorithm employs lightweight 3D cost volume and channel dimension feature enhancement, significantly reducing computational complexity while ensuring accuracy, achieving high inference speed suitable for real-time embedded scenarios.

To demonstrate the superiority of LightStereo, we compared four versions (s, m, l, h) trained on the SceneFlow dataset with traditional algorithms on KITTI 2015, as shown in TABLE II.

TABLE II. PERFORMANCE COMPARISON OF DIFFERENT ALGORITHMS ON KITTI 2015

Algorithm	D1-error (%)	EPE	3PE (%)	FPS
BM	41.27	17.063	42.32	18.9
SGBM	23.10	7.483	24.21	5.1
LightStereo-S	11.67	1.982	12.04	9.4
LightStereo-M	7.74	1.598	8.05	6.8
LightStereo-L	7.55	1.588	7.72	3.2
LightStereo-H	8.11	1.565	7.52	2.5

The results show that all LightStereo versions significantly outperform traditional algorithms in accuracy metrics. Considering accuracy requirements, the LightStereo-M version was selected as the base model. A pair of left and right camera images from the KITTI 2015 dataset was selected for stereo matching using three algorithms to obtain disparity maps as shown in Fig. 3.

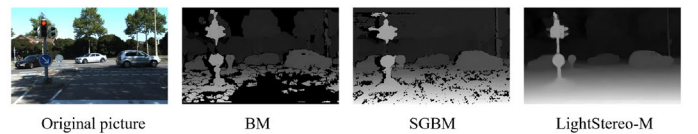


Fig. 3. Parallax maps of different algorithms

C. Dual-Model TensorRT Quantization

The hardware platform of this system is NVIDIA Jetson Orin NX 8GB, with software environment based on JetPack 6.0 (L4T 36.3). All model optimization and performance testing were completed on this platform. To achieve real-time processing on embedded platforms, TensorRT FP16 quantization is employed, retaining FP16 activation values, dynamic range calibration for weights, layer fusion, and kernel auto-tuning^[12]. Compared to INT8 quantization which may cause significant accuracy degradation, FP16 quantization achieves substantial inference speed improvement with minimal accuracy loss. Quantitative Results Of Object Detection Model

First, FP16 quantization was applied to the object detection model. Experimental results show that inference speed improved from 75.31 FPS to 238.09 FPS, achieving approximately 3.2× acceleration with only minor accuracy degradation as shown in TABLE III.

TABLE III. QUANTITATIVE RESULTS OF OBJECT DETECTION MODEL

Algorithm	mAP50	mAP75	mAP50-95	FPS
CSP-AWMSA	0.8283	0.6302	0.5755	75.31
TensorRT-FP16	0.8272	0.6290	0.5736	238.09

Similarly, the stereo matching model achieved frame rate improvement from 6.8 FPS to 23.1 FPS after FP16 quantization, as shown in TABLE IV. Despite minor accuracy loss, for visual impairment assistance tasks, this over 3× real-time improvement provides smoother and more reliable depth perception.

TABLE IV. QUANTITATIVE RESULTS OF STEREO MATCHING MODEL

Algorithm	D1-error (%)	EPE	3PE (%)	FPS
LightStereo-M	7.74	1.598	8.05	6.8
TensorRT-FP16	11.01	1.713	11.28	23.1

III. EMBEDDED PLATFORM DEPLOYMENT AND TEST

A. Hardware Selection

The system adopts a three-tier modular architecture, as shown in Fig. 4, designed to balance high performance, low power consumption, and portability to meet the practical requirements of a visually impaired assistance device.

- Perception Layer: Uses HBVCAM-4M42HD-2V22 industrial-grade binocular camera module with 120mm baseline distance, $f = 3\text{mm}$, $\text{FOV} = 80^\circ$ distortion-free manual zoom capability, calibrated with MATLAB achieving reprojection error $< 0.1\text{px}$.
- Computation Layer: Uses NVIDIA Jetson Orin NX (8GB LPDDR5) development board, operating in 15W Max-P mode (CPU + GPU combined power consumption).
- Interaction Layer: Uses SSS1629 voice module, calling iFlytek TTS text-to-speech API. Detected obstacle categories and distance information are converted to clear voice prompts in real-time, providing intuitive, non-visual environmental information to users.

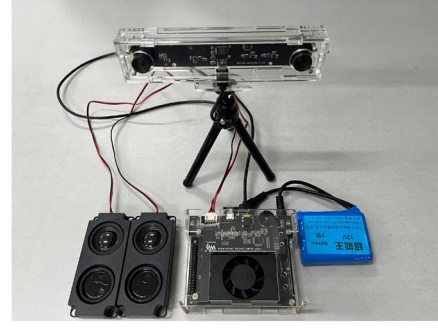


Fig. 4. System hardware composition

B. System Environment and Workflow

The system operates in a software environment built on the NVIDIA JetPack 6.0 development kit. This kit integrates Ubuntu 22.04 operating system, CUDA 12.6, TensorRT 10.0.1, and related drivers and libraries. The main programming language is Python 3.10.

1) Binocular Ranging Principle

The system's distance measurement core is based on classical triangulation principles. When binocular cameras capture synchronized images, the system calculates the disparity d between corresponding points in left and right images through stereo matching. Using calibrated binocular camera parameters, the absolute distance Z of an object can be calculated using the following formula (1):

$$Z = \frac{f \cdot B}{d} \quad (1)$$

where Z is the distance from object to camera, f is the camera focal length, B is the binocular camera baseline, and d is the disparity. This formula indicates that with fixed focal length and baseline, object distance is inversely proportional to disparity.

2) System Workflow

The complete system workflow can be summarized in the following main steps, with the inference flow chart shown in Fig. 5.

- Image Acquisition: The binocular camera synchronously captures raw left and right images.
- Stereo Matching: The left-right image pair is fed into the TensorRT FP16-optimized LightStereo-m model to generate a disparity map.
- Object Detection: Left images are fed into the TensorRT FP16 optimized YOLOv11n model to detect and localize various obstacles in images, outputting their bounding box coordinates.
- Distance Calculation: The system uses YOLOv11n output bounding box coordinates to query pixel values in corresponding regions of the disparity map. By calculating the average disparity of all pixels within the bounding box, the absolute distance of that obstacle is calculated using triangulation formulas.
- Voice Feedback: Detected obstacle categories and calculated distance information are broadcast to users in real-time through the voice module.

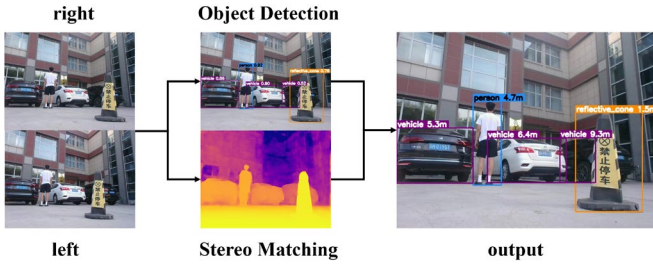


Fig. 5. Reasoning flow chart

C. System Function Testing

1)Dual-Model Fusion Real-Time Test: System functionality was validated on the Jetson Orin NX platform through multi-threaded architecture that parallelizes the two TensorRT FP16 quantized models. Through shared single CUDA context and asynchronous execution using CUDA streams^[13], system performance bottleneck is determined by the more computationally intensive stereo matching module. This parallel optimization strategy ultimately achieved 46.76 milliseconds end-to-end latency from image acquisition to voice output (as shown in TABLE V). Results indicate that the system successfully meets required real-time performance through effective parallel deployment.

TABLE V. STANDALONE AND FUSION INFERENCE LATENCY

Algorithm	Latency (ms)
CSP-AWMSA	4.20
LightStereo-M	43.29
Fusion	46.76

2)Distance Estimation Accuracy Test: Various test objects were placed at different distances from the camera, with ten measurements taken at each position to obtain mean values. Test results are shown in TABLE VI.

TABLE VI. DISTANCE ESTIMATION RESULTS

Truth Distance (m)	Mean Measured Distance (m)	Absolute Error (m)	Relative Error (%)
1.0	1.02	0.02	2.0
2.0	2.09	0.09	4.5
3.0	3.14	0.14	4.7
4.0	4.25	0.25	6.3
5.0	5.43	0.43	8.6
6.0	6.58	0.58	9.7
7.0	7.71	0.71	10.1
8.0	8.92	0.92	11.5
9.0	10.15	1.15	12.8
10.0	11.41	1.41	14.1

As shown in Table VI, the system achieves ideal balance between accuracy and sensing range. At close distances (0-4 meters), its mean absolute error is below 0.25 meters (relative error < 6.3%), which is crucial for active obstacle avoidance. Due to inherent characteristics of stereo vision, errors increase with distance, but the system can still provide reliable distance

trends and warnings within 10 meters. Overall, the system performs effectively and reliably in obstacle detection and distance measurement, basically meeting the core safety requirements of visual impairment assistance systems.

IV. CONCLUSIONS

This study presents a real-time obstacle perception and distance estimation system based on binocular vision for visually impaired individuals. The system integrates an improved YOLOv11n model with CSP-AWMSA module achieving 82.83% mAP@0.5 and a lightweight LightStereo network for depth estimation. Through TensorRT FP16 quantization, the detection and stereo matching models achieve 3.2× and 3.4× speedup respectively. Deployed on NVIDIA Jetson Orin NX, the system demonstrates 46.76 ms end-to-end latency with < 6.3% relative error within the critical 4-meter range. This research successfully validates a practical navigation assistance solution meeting real-time and accuracy requirement while providing a valuable framework for deploying advanced computer vision models on embedded assistive devices.

REFERENCES

- [1] S. Khan, S. Nazir, and H. U. Khan, "Analysis of navigation assistants for blind and visually impaired people: a systematic review," *IEEE Access*, vol. 9, pp. 26712-26734, Feb. 2021, doi: 10.1109/ACCESS.2021.3052415.
- [2] W. A. Al-Muftil, U. Hashim, A. Fraihat, and T. Adam, "Studying on design and simulation of electronic IR guide for blindness," in *AIP Conference Proceedings*, vol. 1775, no. 1, pp. 030103, Oct. 2016.
- [3] L. Wang, Y. Xu, R. Zhu, S. Ding, J. Yu, T. Sun, and B. Liu, "An improved obstacle detection and segmentation method based on VIDAR and binocular vision," *Engineering Letters*, vol. 31, no. 1, Jan. 2023.
- [4] Y. Liu, P. Ou, X. Xu, and J. Sun, "Multi-line structured light binocular vision stereo matching method via coarse-to-fine spatial geometric constraints," *Optics and Laser Technology*, vol. 176, pp. 110950, Aug. 2024.
- [5] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: unified, real-time object detection," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 779-788.
- [6] R. Khanam and M. Hussain, "YOLOv11: an overview of the key architectural enhancements," *arXiv preprint arXiv:2410.17725*, Oct. 2024.
- [7] H. Xia, C. Yao, Y. Tan, and S. Song, "A dataset for the visually impaired walk on the road," *Displays*, vol. 79, pp. 102486, Oct. 2023.
- [8] B. Almatrouk, H. Meng, and R. M. Swash, "Holoscopic elemental-image-based disparity estimation using multi-scale, multi-window semi-global block matching," *Applied Sciences*, vol. 14, no. 8, pp. 3335, Apr. 2024.
- [9] L. Lipson, Z. Teed, and J. Deng, "RAFT-stereo: multilevel recurrent field transforms for stereo matching," in *Proc. 2021 Int. Conf. 3D Vision (3DV)*, London, UK, Dec. 2021, pp. 218-227.
- [10] G. Xu, X. Wang, X. Ding, and X. Yang, "Iterative geometry encoding volume for stereo matching," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, Vancouver, BC, Canada, Jun. 2023, pp. 21919-21928.
- [11] X. Guo, C. Zhang, Y. Zhang, W. Zheng, D. Nie, M. Poggi, and L. Chen, "LightStereo: channel boost is all you need for efficient 2D cost aggregation," *arXiv preprint arXiv:2406.19833*, Jun. 2024.
- [12] E. Jeong, J. Kim, and S. Ha, "TensorRT-based framework and optimization methodology for deep learning inference on Jetson boards," *ACM Trans. Embed. Comput. Syst.*, vol. 21, no. 5, art. 51, Oct. 2022.
- [13] W. Pang, X. Luo, K. Chen, D. Ji, L. Qiao, and W. Yi, "Efficient CUDA stream management for multi-DNN real-time inference on embedded GPUs," *Journal of Systems Architecture*, vol. 139, pp. 102888, May 2023.